Unit 9, Lesson 5: Interpreting Box Plots

Benchmark

MA.6.DP.1.3 Given a box plot within a real-world context, determine the minimum, the lower quartile, the median, the upper quartile, and the maximum. Use this summary of the data to describe the spread and distribution of the data.

Learning Targets

- ☐ I can determine the minimum, lower quartile, median, upper quartile, and maximum from a box plot.
- ☐ I can determine the interquartile range from a box plot.
- ☐ I can describe the spread and distribution of data from a box plot.

9.5.1 Warm-Up: Which One Doesn't Belong?

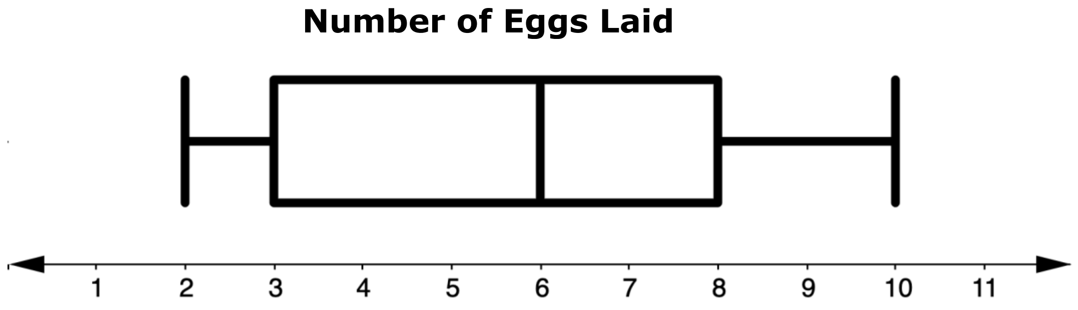
Examine the four numbers.

33%	$\frac{1}{3}$
$\frac{5}{3}$	$0.\overline{6}$

1. Which of these numbers does not belong? Explain your thinking.

9.5.2 Exploration: Exploring Box Plots

1. A farmer recorded the number of eggs her hens laid in one week. The data set was the total number of eggs each of her 53 hens laid during the week. Below is a box plot representing the number of eggs laid by each hen during the week.



With your partner, use the box plot to complete the statements.

a. The smallest number of eggs laid by the hens is _______.

b. The largest number of eggs laid by the hens is _______.

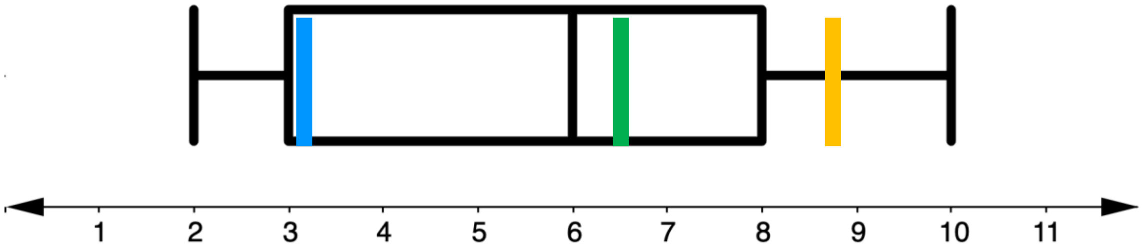
c. Find another pair of partners and compare the smallest and largest number of eggs laid. Discuss how you determined the values from the box plot. If your values are different, come to a consensus on the smallest and largest number of eggs laid by the hen, and circle these values on the box plot.

d. Label the range for the number of eggs laid on the box plot.

e. With your partner, conjecture the median number of eggs laid. Label this number as “median” on the box plot.

f. The first quartile is highlighted in blue, the second quartile is highlighted in green, and the third quartile is highlighted in yellow.

Number of Eggs Laid



If the second quartile is the median, M, write a conjecture with your partner about how the first and third quartiles are each determined.

2. Work with your partner to complete the statement.

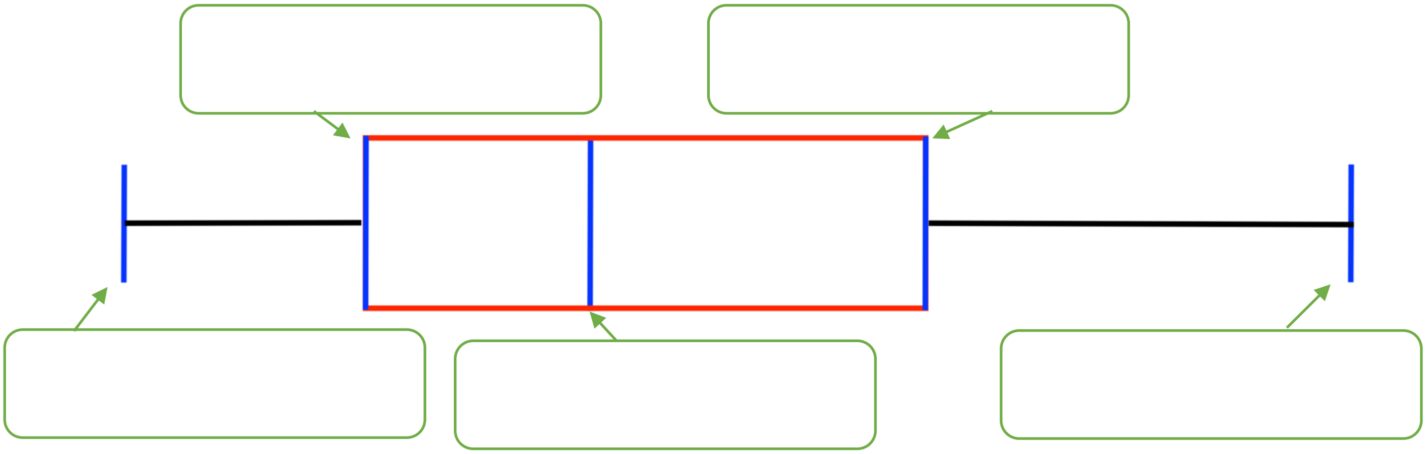
A ***box plot*** divides data into ________ equal parts that each contain roughly 25% of data from a given data set.

3. Explain why you chose the value you used to complete the statement.

9.5.3 Guided Instruction: Interpreting Box Plots

A ***box plot*** is a plot displaying the spread or distribution of a data set using a five-number summary: the minimum, lower quartile (first quartile), median, upper quartile (third quartile), and maximum. It is also called a box-and-whisker plot.

1. Label the maximum, minimum, median, first quartile, and third quartile on the box plot.



2. Complete the statement.

The ***first quartile (Q_1)*** is the middle value of the data

that are
- ○ less than
- ○ greater than
- ○ equal to

the median.

The ***third quartile (Q_3)*** is the middle value of the data

that are
- ○ less than
- ○ greater than
- ○ equal to

the median.

For a data set with median, M, the first ***quartile*** is the median of the data values less than M, and the third ***quartile*** is the median of the data values greater than M. The second ***quartile*** is the median, M.

3. Complete the statement.

The measure of central tendency used to create a box plot is the
- ○ mean,
- ○ median,
- ○ mode,

which divides the data set in half.

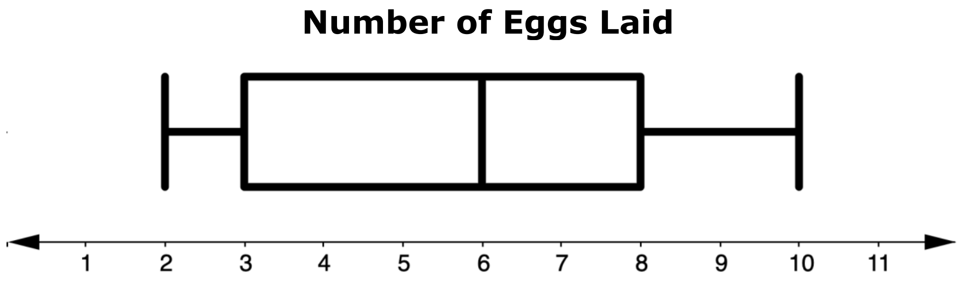
A number line often accompanies a box plot and is used to determine the values associated with each part of the plot.

Another measure of variability like ***range***, the ***interquartile range (IQR)***, can be found by determining the difference in the third quartile from the first quartile.

A measure of variation in a set of numerical data, the ***interquartile range*** is the distance between the first and third quartiles of the data set.

IQR = (Q_3) - (Q_1)

4. Revisit the box plot of the eggs laid by a farmer's hens.

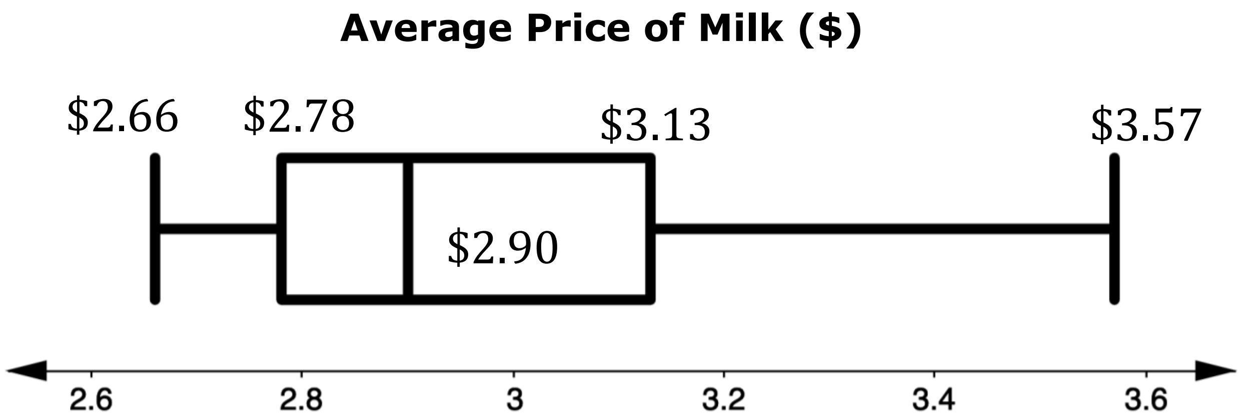


a. What is the interquartile range (IQR) for the box plot of the context?

b. Label the IQR on the box plot.

9.5.4 Guided Practice: Interpreting Box Plots

1. The average price of milk, in dollars ($), for each month from April 2000 to November 2006 was used to create the box plot shown.



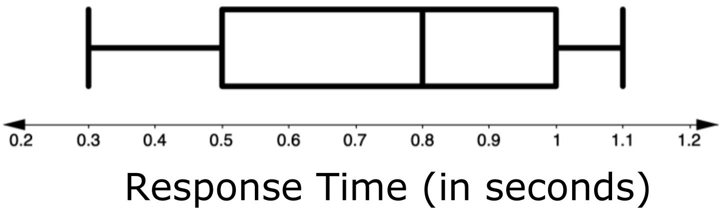
a. What is the range of the data?

b. Determine the interquartile range.

c. Does the average cost of milk from April 2000 to November 2006 vary a lot or a little? Explain your reasoning.

9.5.5 Try It: Interpreting Box Plots

1. The box plot displays data on the response times of 100 mice when seeing a flash of light.



Complete the statements.

a. The fastest response time is _______ seconds.

b. The slowest response time is _______ seconds.

c. The middle response time is _________ seconds.

d. The lower quartile is _____________ seconds.

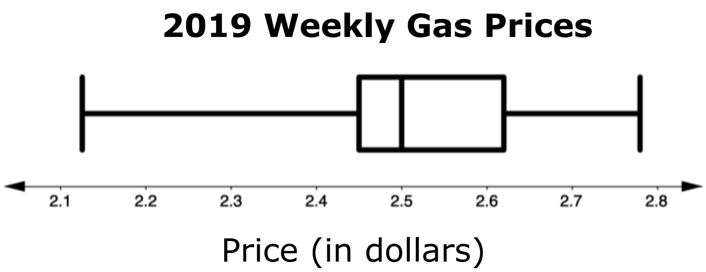
e. The upper quartile is _____________ seconds.

f. The range is ____________ seconds.

g. The interquartile range is ____________ seconds.

2. With your partner, use the values to write a description of the spread and distribution of response times.

3. The U.S. Energy Information Administration reports weekly average gas prices for the United States. The box plot represents the weekly average gas prices in 2019.

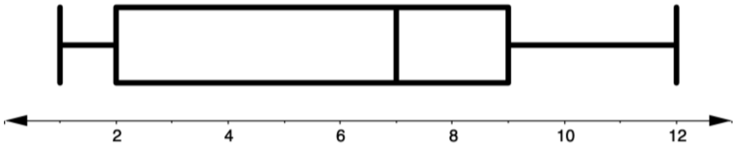


What percentage of weeks had an average gas price of at least $2.50?

9.5.6 Your Turn!

1. Mrs. Henry's 3rd period English class took a spelling test with 12 words on it. A box plot was created from the number of questions students got correct on the test.

Number of Correct Questions



Complete the statements.

a. The lowest score is _______.

b. The highest score is _______.

c. The middle score is _________.

d. The lower quartile is _____________.

e. The upper quartile is _____________.

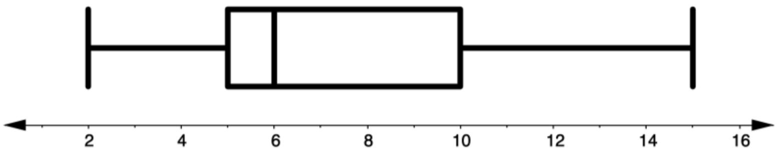
f. The range is ____________.

g. The interquartile range is ___________.

h. Write a description of the spread and distribution of the scores.

2. The librarian at Franklin Middle School surveyed students when they returned from summer break on how many books they had read. She created a box plot to represent the results of her survey.

Number of Books Read



What percentage of students read between:

a. 5 and 10 books over the summer? _________

b. 10 and 15 books over the summer? _________

c. 5 and 6 books over the summer? _________

d. 2 and 10 books over the summer? _________

9.5.7 Wrap-Up: Cool Down

Jamal missed class today and he needs help with what he missed.

1. Write a note to Jamal letting him know what a box plot is and what is it used for. Draw a picture or diagram that may help him better understand.